

Lambert
electromec Ltd

DATA CENTRE AND TELECOM INFRASTRUCTURE



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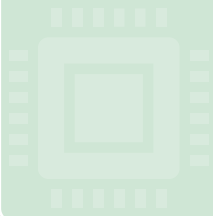
Location

About Us

Lambert Electromec continuously expands its resources and services to meet the changing needs of its customers in a rapidly evolving industry, our focus industries are as follow:



Data Center And Telecom Infrastructure



Building Services

(Offices, Hospitals, Hotels,
Institutions, Factories, Mixed-Use,
Residential, etc.) and Industrial
Plants



Power Plants And substations



MEP For Infrastructural And Enabling Works

(Rail Stations, Stadiums and Ports)



Oil & Gas Modular Refineries



History

Lambert Somec Nig. Ltd . Was created in affiliation with Lambert Somec Inc., a leading Electromechanical construction company from the province of Quebec in Canada

Before separating with its Nigerian partner in 2004, Lambert Somec Inc. passed on its heritage of first-class professionalism, top engineers and its exceptional overall operation system.

In 2007, it then changed its name to Lambert Electromec Nig. Ltd. And transferred its main operations to Lagos.



What We Do



**Building and Factories
Electrical &
Mechanical
Engineering**



**Power Generation
And Distribution**



**Data Center &
Telecom
Infrastructure**



**Oil and Gas, E&I
And Piping**



Maintenance

Industries We Serve

Lambert Electromec continuously expands its resources and services to meet the changing needs of its customers in a rapidly evolving industry, our focus industries are as follow:



Bank
s



Oil & Gas Facilities



Telecom Service



Factories



Healthcar
e



Hotel/ Apartment
Building



Large
Enterprise



Malls



Educational
Institution



New Construction



Offices



Warehouse



Small & Medium
Facilities

Data Center Facility

Lambert Provides Design & Build Data Center Solutions covering the below Passive IT Infrastructure components



Power

- UPS Systems (Stand alone and Modular)
- Panel Boards (Main and Distribution)
- Power Distribution Units
- STS (Static Transfer Switch)



Cooling

- CRAC DX
- LCP
- Chillers
- CRAH



Containment

- Hot Aisle Design
- Cold Aisle Design



Cabinets

- IT Racks
- Network Racks
- Server Racks
- UPS Racks



Environmental Monitoring System

- Temperature Sensors
- Humidity Sensors
- Leak Detector Sensors



Structured Cabling

- Copper Solution
- Fiber Solution



Raised Floor



DCIM

- Data Center Infrastructure Management Software



Civil & Electrical Works

- False Ceiling
- Lighting



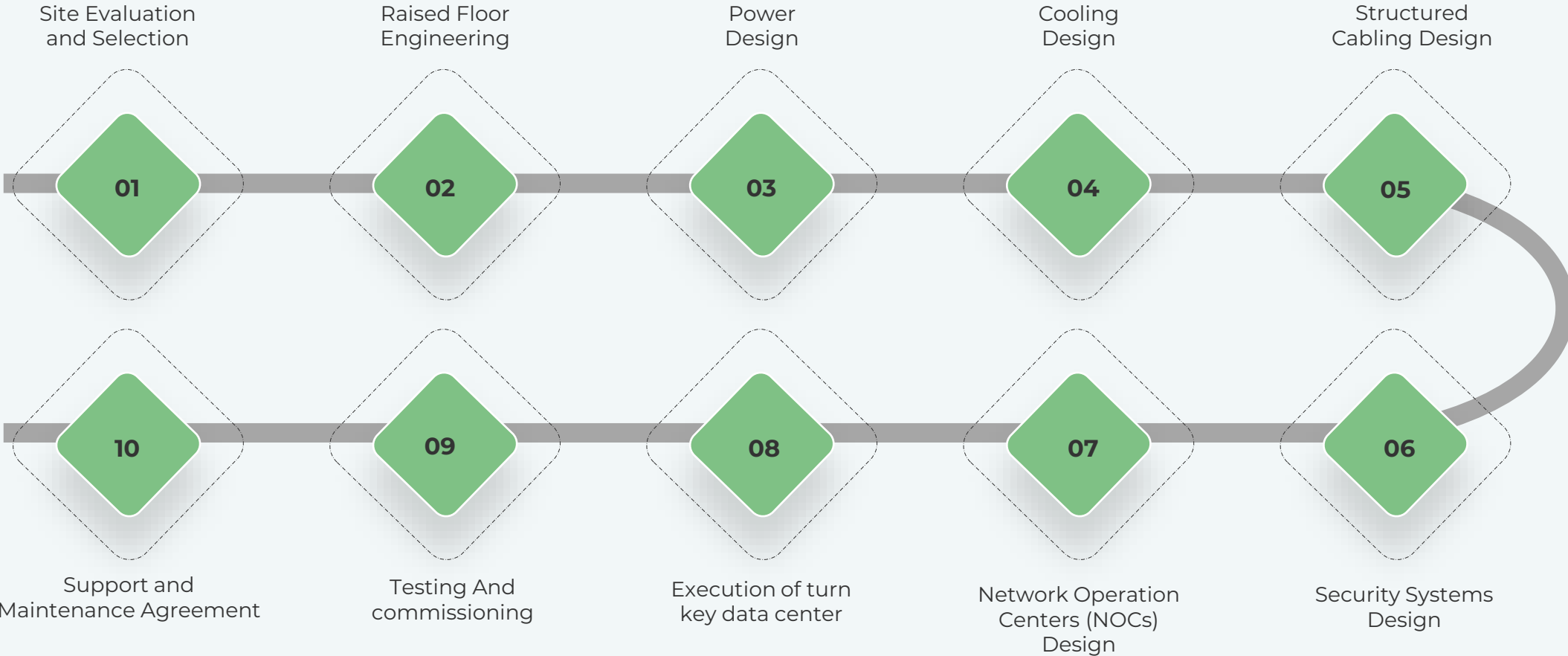
Physical Security

- Access Control
- CCTV
- Intrusion



Fire Fighting and Fire Alarm System

Data Center Facility





Data Center Facility

Design

Our ATD certified engineers will provide Tiering design according to Uptime Institute to meet client's requirements:

Achieve the desired level of site infrastructure performance or uptime (increase the availability).

Reduce the complexity of IT environment management

Improve Productivity and time

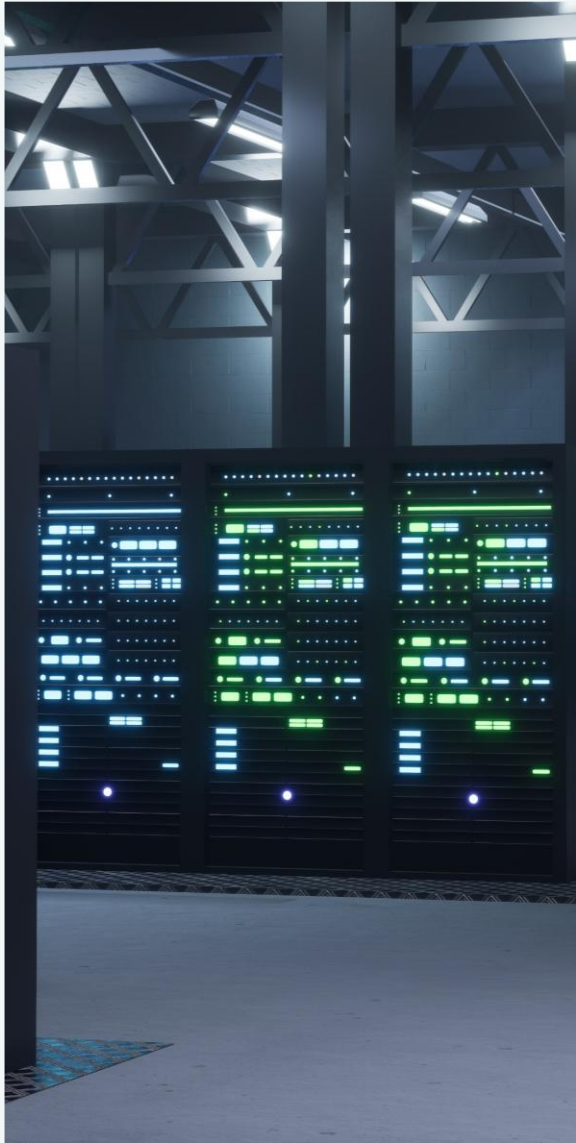
Protect your critical data through integrating data security, backup and recovery strategies that improve efficiencies and reduce upgrade costs

Optimizing data center efficiency to realize significant cost savings

Incorporating virtualization to consolidate resources and greatly reduce IT Costs

Ensuring business continuity in the event of a disaster with comprehensive recovery plans

Data Center Facility



Design Methodology

Power Design

The proposal will be based on the following:

Number of distribution paths
according to the required tier level

Active capacity components needed
to support the IT load

The design covers the selection of

Back up power source
(Generators)

Required utility
supply load

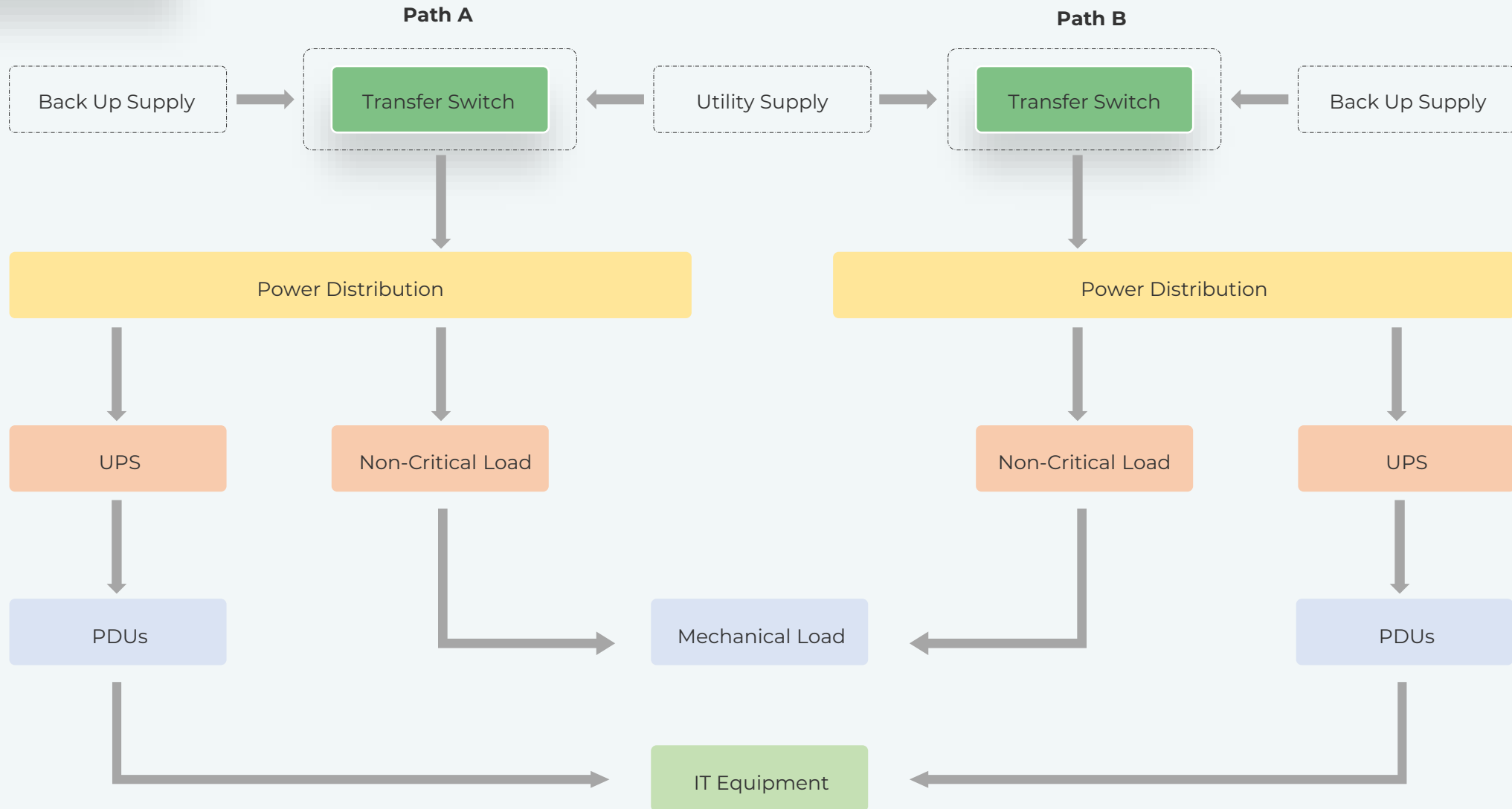
Main distribution panel
boards (paths as per
required tier level)

UPS system

Power distribution units
feeding the critical loads

Static transfer switches
providing the redundancy
for single powered IT loads

Power Design



Cooling Design



Method A

**Room Cooling For low power consumption
per rack (≤ 4 kW)**

CRAC

or

Chilled water supply or DX

Principle

Cold air moves under the raised floor

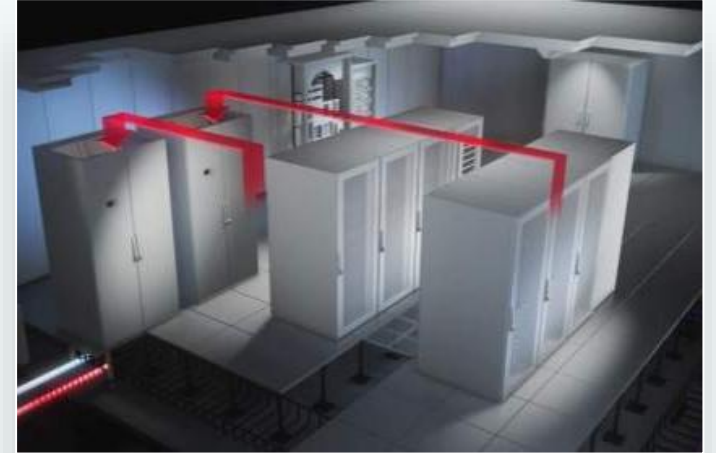
Racks are cooled through perforated tiles and
perforated server doors

Warm exhaust air at rear door

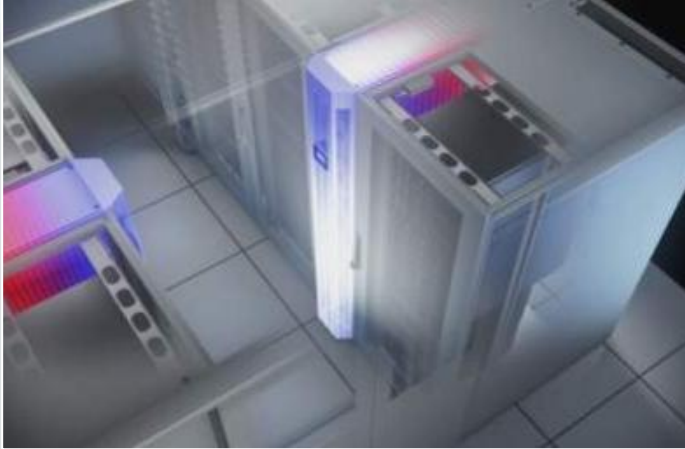
CRAC unit takes in the warm air

A heat exchanger cools down the air

Circulation starts again



Cooling Design



Method B

**In Row Cooling for mid range power
consumption (<10 Kw)**

Heat Exchanger inside rack row
Chilled water or DX

Principle

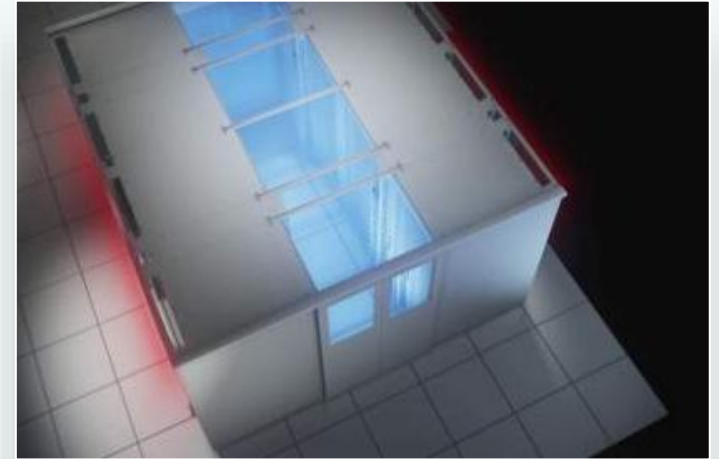
Cold air blown in the cold aisle

Perforated server doors

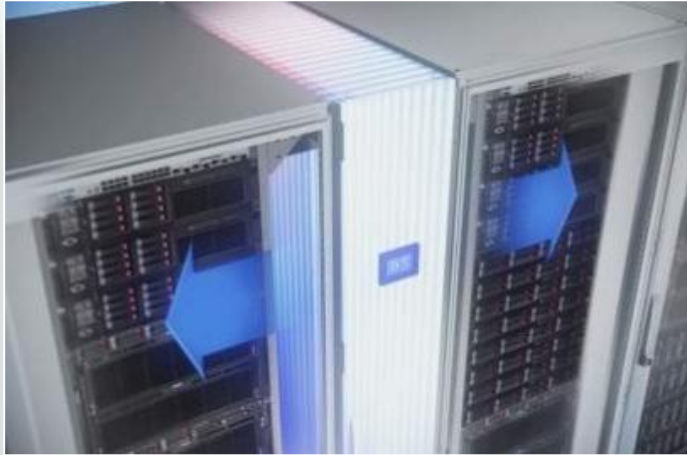
Warm exhaust air at rear door

Heat exchanger takes in warm air

Circulation re-starts



Cooling Design



Method C

In Rack Cooling for High power consumption (>10 kW)

Heat exchanger in row of racks
Dedicated to one or two racks

Principle

Cold air blown behind glass door in front of the servers

Warm exhaust air at backside of servers but inside rack

Exhaust air will be taken in by heat exchanger
Circulation re-starts



Other Components Design

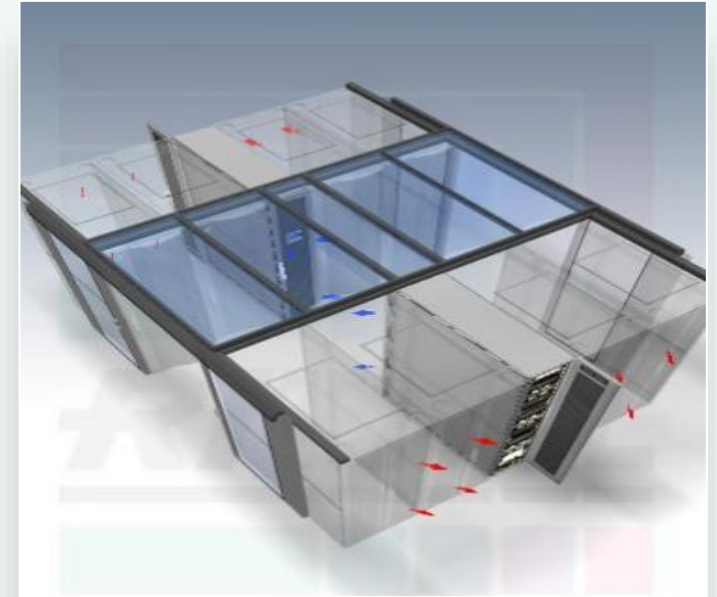
Aisle Containment - Methods

A cold-aisle containment system (CACS) encloses the cold aisle, allowing the rest of the data center to become a large hot-air return plenum. By containing the cold aisle, the hot and cold air streams are separated. Note that this containment method requires that the rows of racks be set up in a consistent hot-aisle / cold-aisle arrangement

A hot-aisle containment system (HACS) encloses the hot aisle to collect the IT equipment's hot exhaust air, allowing the rest of the room to become a large cold-air supply plenum. By containing the hot aisle, the hot and cold air streams are separated. Note that this containment method requires that the rows of racks be set up in a consistent hot-aisle / cold-aisle arrangement. Note that there are two basic methods for containing the hot aisle – row-cooled hot aisle containment and ducted hot aisle containment.

Regardless of the type of containment system, people still need to work inside a data center. This uncontained area must be kept at a reasonable temperature so as not to violate health and human safety regulations for exceeding wet-bulb globe temperature (WBGT)⁶. Note the following difference in the uncontained area.

- With cold-aisle containment, the uncontained area becomes the same temperature as the hot aisle.
- With hot-aisle containment, the uncontained area becomes the same temperature as the cold aisle.



Environmental monitoring System

Leak Detectors

Temperature and humidity sensors

Doors Status

Other Components Design



DCIM Monitoring System

Structured Cabling Design

Copper and Fiber Design by certified team

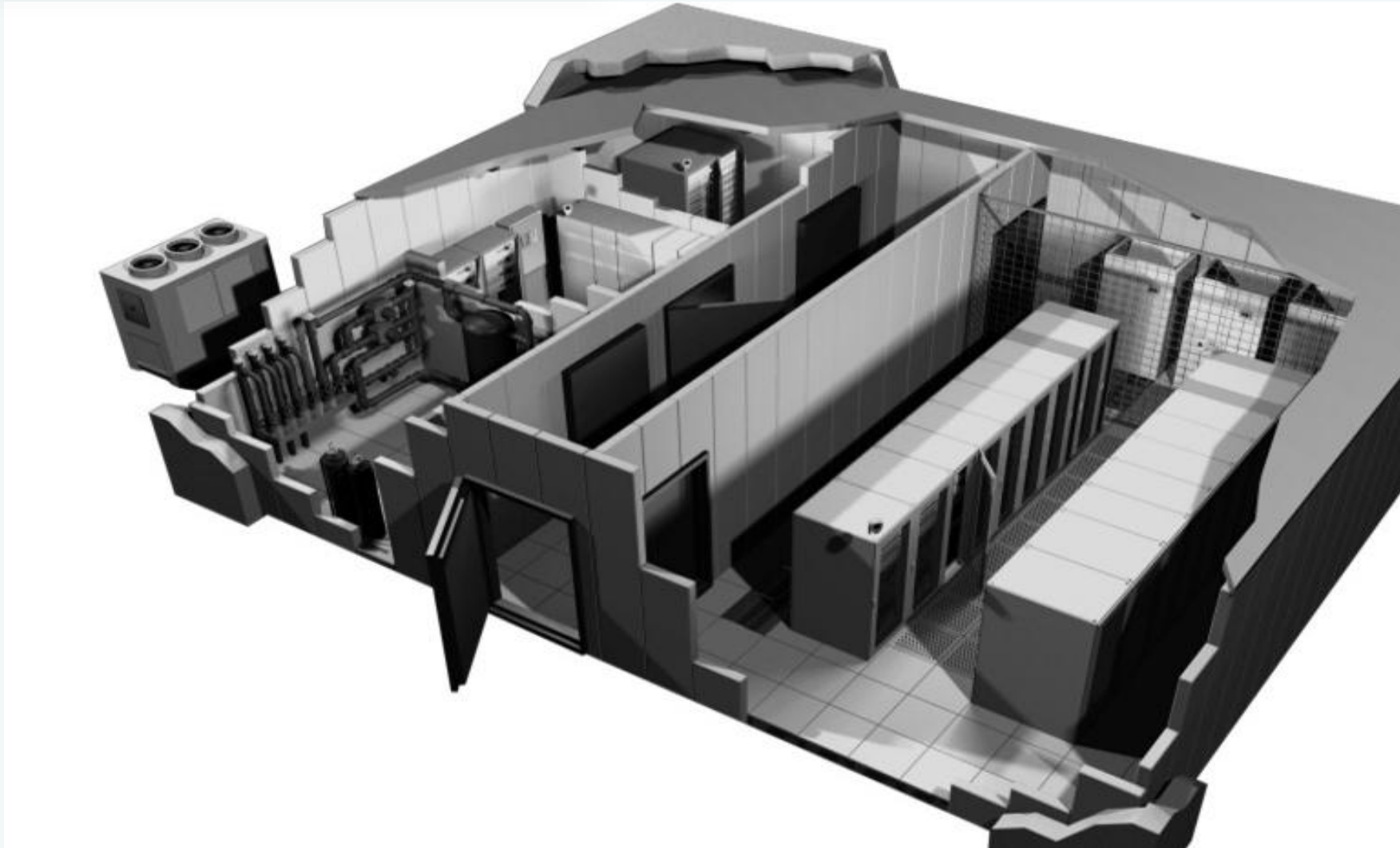
Indoor and outdoor Design

Horizontal and Vertical connectivity

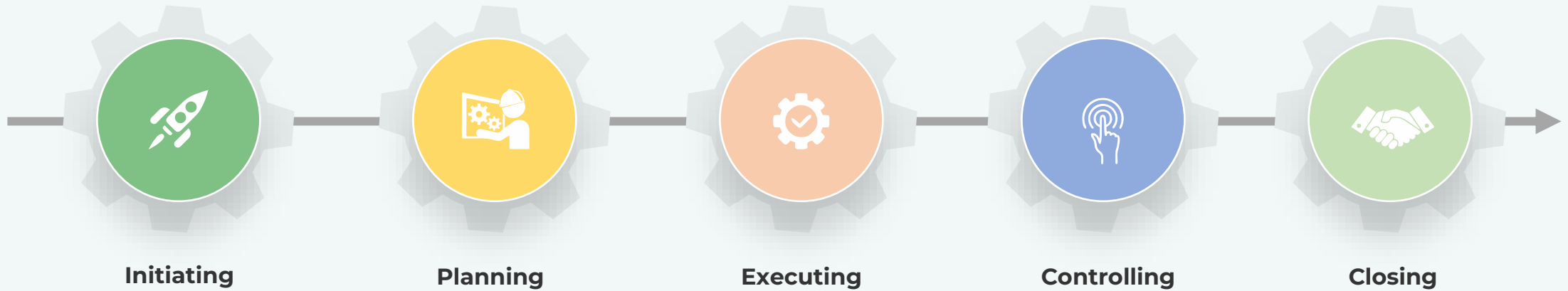
Latest technology proposal (Cat7, Cat8, Pre-terminated Fiber Cables...)



Data Center Facility Execution Methodology



Project Management Methodology



Team Responsible For Execution

Dedicated Project
Manager

Data Center Certified
Engineers

Certified team leaders
and technicians



Initiating Phase

This phase Includes Bid Qualification/Proposal phases from the business lifecycle. PM engagement is authorized at this level to provide project management related input

- 1 What the project will create / implement
- 2 Study Customer RFP / Requirements
- 3 Business Risk Assessment
- 4 Opportunity Feasibility
- 5 High level Design and Implementation Planning



Planning Phase

This phase focuses on scoping and planning the project. Planning is split into two phases

Pre-Contract

1

What the project will create / implement

2

Study Customer RFP / Requirements

3

Business Risk Assessment

4

Opportunity Feasibility

5

High level Design and Implementation Planning

Post-Contract

1

Resources assignment / allocation

2

Project kick off meeting

3

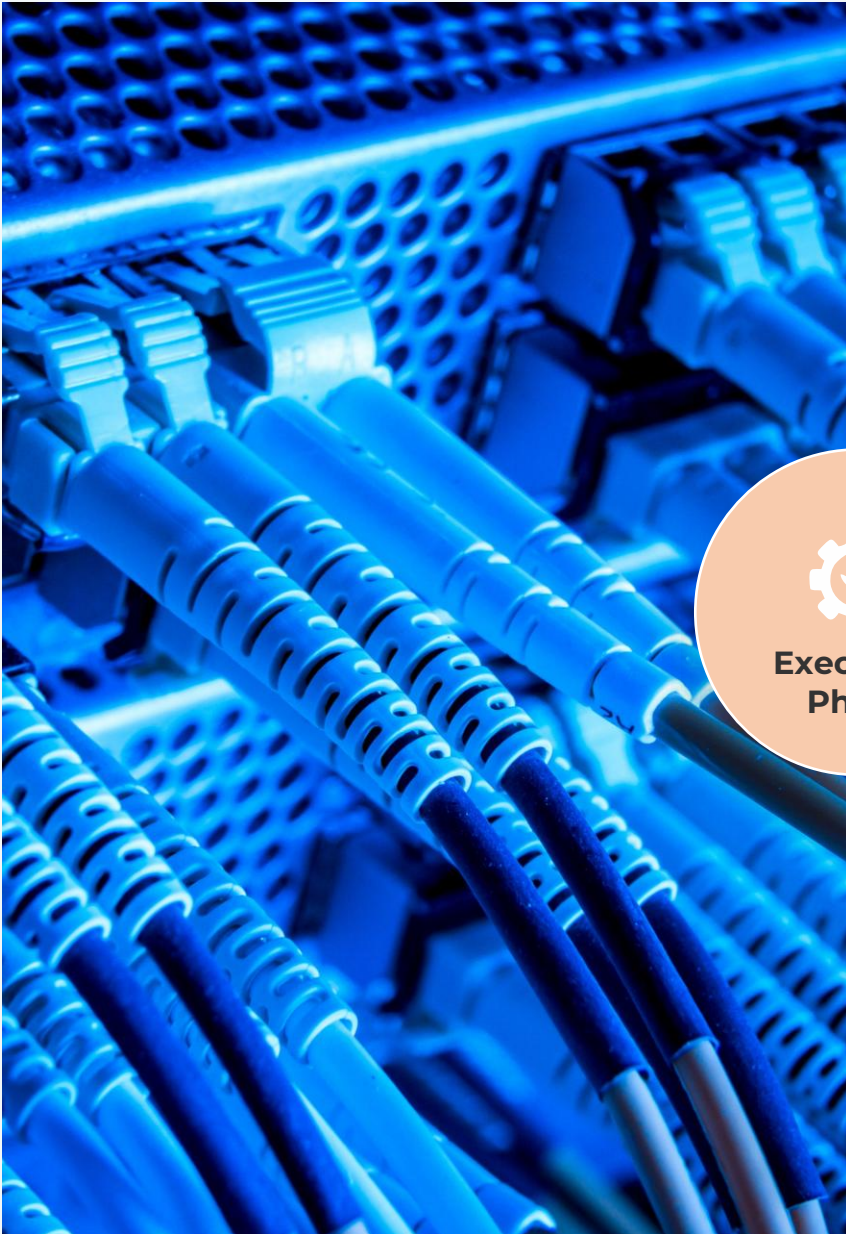
Workshops for training and implementation requirement gathering

4

Project Definition Document

5

Planning for the execution, design & acceptance documents preparations



Executing Phase

This Phase includes the delivery of a complete accepted solution that meets fully the project requirements

1

Deliverables Execution / Submittals

2

Design Documents Compilation (LLD, Migration Plan, NRFU).

3

Shop Drawings Preparations, if required

4

Installation / Staging, configuration, swapping, commissioning, testing of equipment

6

Communication Management

7

Scope Verification / Inspection

8

Obtain Formal Acceptance on Deliverables



This phase runs throughout the life of the project to keep stakeholders informed and project under the right control

**Controlling
Phase**

1

Quality Control

2

Change Control

3

Progress Review

4

Risk / Issues Review and Updates



Closing Phase

This phase ensures the project has a distinct end-point and there is a definite transfer of ownership of the project outputs to the customer and/or support organization

1

Formal Acceptance of all Project Deliverables

2

Handover Documentation

3

Post Implementation Training (if required).

4

Lessons Learnt

5

Commissioning

6

Validation of design concept and desired reliability level

7

Validation of intended systems operation as an integrated whole

8

Commissioning Deliverables

9

Factory Witness Tests

10

Construction Observation Comments

11

Pre-Functional Checklists

12

Functional Performance Tests

13

Written Report Documenting Results

Mastering MEP Integration in Data Centers



The Rack Center

Lagos - Nigeria

Rack Centre Phase 2 (LGS2) is a Tier III carrier-neutral data centre expansion located in Ikeja, Lagos. The facility is designed to deliver 13.5MW of IT power with 3,240sqm of lettable white space, positioning it as one of the largest and most advanced colocation facilities in West Africa.

Configuration

Tier 3 Data Center

Other Projects in Data Centers



GT Bank Data Center

Lagos - Nigeria

Scope

Mechanical, Electrical
& Plumbing (MEP)
Works

Client

GT Bank

Configuration

Tier Level 3, 75 Racks, 6PDU Racks



TOTALEnergies Data Centres & Telecom Towers

Port Harcourt - Nigeria

Scope

Mechanical, Electrical
& Plumbing (MEP)
Works

Client

Total Elf

Configuration

Tier Level 3



TOTALEnergies Data Centers

Lagos - Nigeria

Scope

Mechanical, Electrical
& Plumbing (MEP)
Works

Client

Total – Eko Hotel

Configuration

2 Nos Tier 3 Data Centers (30 racks in total).

Other Projects in Data Centers



The Rack Center - Phase I

Lagos - Nigeria

Scope

Mechanical, Electrical
& Plumbing (MEP)
Works

Client

The Rack Center

Configuration

Tier Level 3



Access Bank Corporate Head Office

Lagos - Nigeria

Scope

Mechanical, Electrical
& Plumbing (MEP)
Works

Client

Access Bank

Configuration

18 Racks / 12 Data Rooms



NCDMB HQ Office Complex

Bayelsa - Nigeria

Scope

Mechanical, Electrical
& Plumbing (MEP)
Works

Client

Nigeria Content
Development & Monitoring
Board (NCDMB)

Configuration

22 racks / data rooms

Design Only Projects - Data Centers



Delta Regional Control Center (DRCC)
Egypt

Configuration

Tier 3 Data Center

32 Racks

Data Center with control room for
the ministry of electricity.



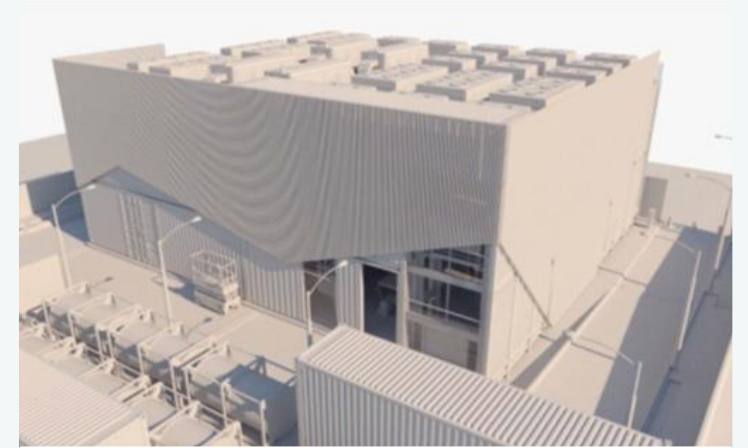
MTN Data Center
Lagos - Nigeria

Configuration

Tier 3 Data Center

106 Racks

Data Center for MTN



Zain Kuwait, SSB Phase 2 Project
Kuwait

Configuration

Tier 3 Data Center

226 Racks

Data Center for Zain

Sustainability Featured Projects

Lambert Electromec is vastly experience in executing sustainability projects, our policies align fully with the sustainability goals and we have showcased that in certified projects in LEED, Green Star, and EDGE. Below are some of our certified projects:



Alpha One Tower
Lagos - Nigeria



IFC World Bank Group Office
Accra - Ghana



Port of Tema Expansion
Accra - Ghana



Ivoire Trade Center
Abidjan – Cote d'Ivoire



Heritage Place
Lagos - Nigeria



Ghana National Petroleum Corporation (GNPC)
Takoradi - Ghana



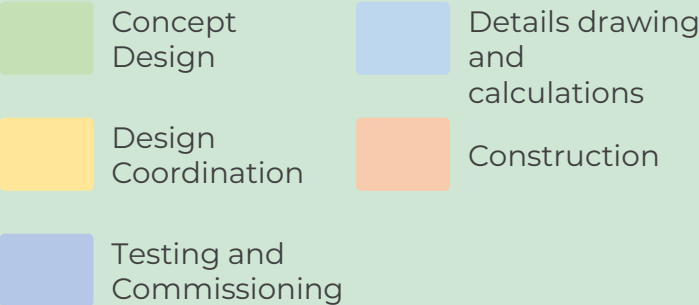
Stanbic IBTC New HQ
Lagos - Nigeria



One Airport Square
Accra - Ghana

Management And Capabilities

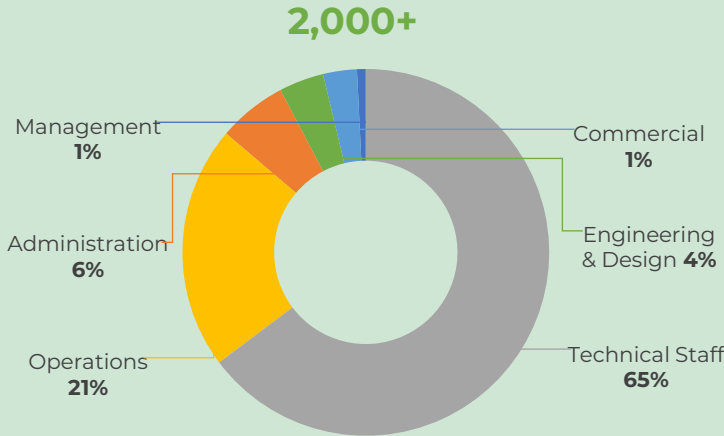
Design & Engineering Capabilities



Health, Safety & Environment

Lambert Electromec has a Health, Safety, and Environment (HSE) policy that conforms to international standards. It is totally committed to the concept of building safety into the design of high technology installations and plants

Workforce



Maintenance, After Sales & Supports

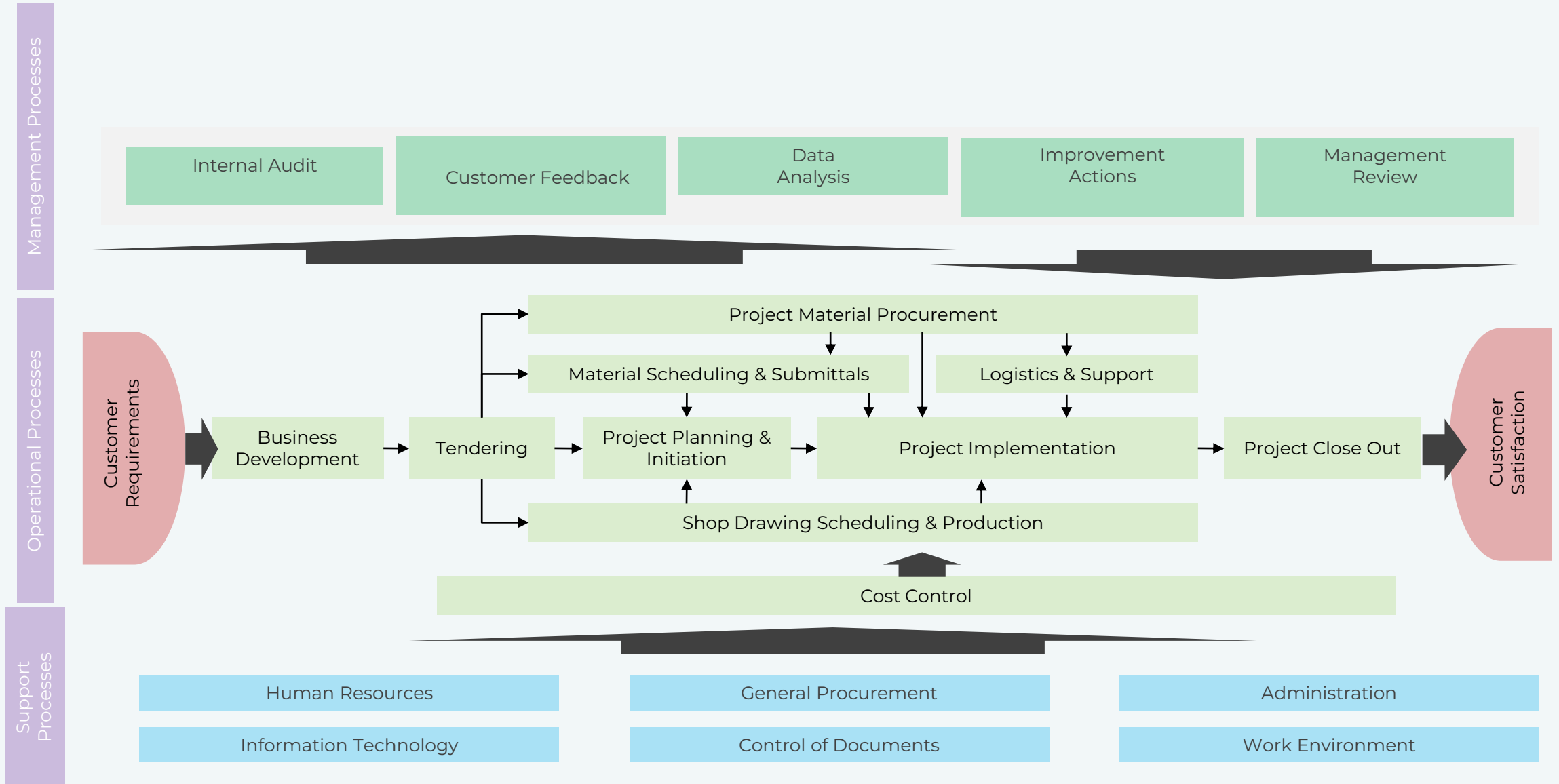


Lambert Electromec provides a complete and comprehensive solution to all its clients' building maintenance requirements, through its subsidiary LEM Maintenance

Quality Management System



Process Map Architecture



Location

CÔTE D'IVOIRE



Abidjan

Cocody Danga Villa,
N°6 – Rue des Cannas
26 BP 13777 Abidjan 26
– Côte d'Ivoire
Cell: +225 05 64 02 37 23

GHANA



Accra

Plot No. 8, 4TH Circular Street
6th link close, Cantonments
Accra East District, Ghana.
Tel: +233 30 224 2603,
+233 59 380 44 01



NIGERIA



Lagos

21, Ahmed Onibudo Street,
Ground floor, Left Wing,
Victoria Island, Lagos.
Tel: +234 91 393 827 69

Abuja

38 TOS Benson
Crescent, Utako
Abuja, FCT.

Port Harcourt

48 Ordinance Road,
Trans Amadi layout,
River State, Nigeria.
P.O. Box 6469, Port Harcourt.
Tel: +234 (84) 487 844/232 501

Thank You

